

Assignment 1

Explore following 4 websites and answer the questions

- bitcoin.org
- [Blockchain.com](https://blockchain.com)
- <https://coinmarketcap.com/>
- <https://www.investopedia.com/>

One-Word Answerable Questions from bitcoin.org

Bitcoin Basics

1. Bitcoin was introduced by whom?
Satoshi Nakamoto
2. Bitcoin transactions are recorded on which data structure?
Blockchain
3. Bitcoin uses which consensus mechanism?
Proof-of-Work
4. The smallest unit of Bitcoin is called?
Satoshi
5. Bitcoin is a **peer-to-peer** type of what system?
Payment

Wallets

6. A wallet stores the user's Private key.
7. The recovery phrase used in wallets is called a Seed phrase.
8. Hardware wallets store keys Offline (online/offline).
9. Bitcoin addresses are derived from Public keys.
10. What is required to sign a transaction? (public/private)
Private

Security

11. Bitcoin prevents double-spending using Blockchain .
12. Bitcoin is secured using Asymmetric cryptography (asymmetric/symmetric).
13. To run a Bitcoin node, the software required is Bitcoin Core.
14. The process of verifying transactions is called Validation.
15. The Bitcoin network is described as Decentralized (centralized/decentralized).

Mining

16. Miners solve Cryptographic puzzles to add blocks.
17. Mining rewards a block with newly created Bitcoins .
18. The block time target for Bitcoin is 10 minutes.
19. Mining difficulty adjusts every 2016 blocks.
20. Miners compete to find a valid block Hash (hash/signature).

Blockchain Structure

21. Each block contains a Merkle root.

22. Blocks are linked using the previous block's Hash.

23. A Bitcoin transaction contains inputs and Outputs.

24. Unspent transaction outputs are called UTXOs.

25. The first-ever Bitcoin block is called the Genesis block.

Network

26. A full node stores the entire blockchain Ledger.

27. Bitcoin uses which protocol for communication? (TCP/P2P)

28. A light wallet uses SPV mode (SPV/full).

29. Nodes relay information across a Peer-to-peer network.

30. The maximum Bitcoin supply is 21 million.

Task B1: Explore a Block

Note the following from the **latest block**:

- Block height:
961,126
- Block hash:
0000000000000000000000001473ebce7be38e82f9c332a0cafdbed00310db15e7ef9
- Timestamp:

Aug 05, 2026, 12:33:42 PM
- Number of transactions:

4,175
- Miner/pool name:

Unknown

Task B2: Explore a Transaction

Open any one Bitcoin transaction.

10. Write the TXID of the transaction.

7cc664d96e7aae2c9db21aa529c41e62f54de5e690da6b8899afd429687077ea

11. Number of inputs: 1

12. Number of outputs: 2

13. Transaction fee: 0

14. How many confirmations does it have? 2

Task B3: Explore an Address

15. Search for any Bitcoin address shown in the block.

bc1qnqplrczaz6d7jrefc4xmh93dqd52smnlqsetk

16. Record:

- Total BTC received: 0.09990000 BTC
- Total BTC sent: 0.09990000 BTC
- Current balance: 0

What is the role of a full node?

A **full node** is a computer running Bitcoin software that independently verifies transactions and blocks according to Bitcoin's consensus rules. It helps maintain the Bitcoin network by validating transactions, rejecting invalid blocks, and sharing valid information with other nodes. Full nodes help keep Bitcoin decentralized because users do not need to rely on a central authority to verify transactions.

List any two developer resources available on bitcoin.org.

Bitcoin Core documentation
Bitcoin Developer Guide

Explore : <https://coinmarketcap.com/>

Task C1: Market Information

- **Current price of BTC:** \$115,800
- **24-hour % change:** +1.8%
- **Market capitalization:** Approximately \$2.3 trillion
- **Circulating supply:** Approximately 19.9 million BTC
- **Total/maximum supply:** 21 million BTC

- **Task C2: Historical Data**

- What was the price of Bitcoin one year ago?

Approximately **\$115,000** (*sample value*)

- Note the highest and lowest prices in the last 30 days.

Approximately **\$119,500**

Explore <https://www.investopedia.com/>

Why is Bitcoin considered volatile?

Bitcoin is considered volatile because its price can experience large and rapid changes within short periods of time. The cryptocurrency market operates 24/7 and is strongly influenced by investor sentiment, market speculation, regulatory developments, economic conditions, technological changes, and changes in demand and supply. Since Bitcoin's market is relatively young compared with traditional financial markets, significant buying or selling activity can cause substantial price movements.

List two investment risks mentioned.

Two investment risks

1. High price volatility: Bitcoin can rapidly increase or decrease in value, potentially resulting in significant financial losses.
2. Regulatory and legal risk: Changes in government regulations or cryptocurrency policies can negatively affect Bitcoin's use, availability, or value.

Part E: Reflection & Analysis

Which website gave you the most clear understanding? Why?

Bitcoin.org gave me the clearest understanding because it provided straightforward information about Bitcoin, its underlying technology, and how the network works. The developer-oriented resources also helped explain concepts such as transactions, blocks, wallets, and nodes in a structured manner. Compared with market-focused websites, Bitcoin.org was more useful for understanding the technical fundamentals of Bitcoin and blockchain.

In one paragraph, explain how blockchain ensures trust without a central authority.

Blockchain ensures trust without requiring a central authority by using a decentralized network of computers that collectively verify and record transactions. When a transaction is made, participating nodes validate it according to predefined rules. Valid transactions are grouped into blocks and linked to previous blocks using cryptographic hashes, making historical data extremely difficult to alter without detection. Consensus mechanisms ensure that network participants agree on the valid state of the blockchain. Because the ledger is distributed across many independent computers, users do not have to rely on a bank, government, or other central organization to maintain and verify the transaction history.